# How Much Adaptation Is Enough?
# Drift Geometry and Minimal Calibration in Brain-to-Text BCI

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## Abstract

Brain-to-text BCI decoders must remain usable as neural signals change across recording sessions. We audit how much input adaptation is needed when the official T15 brain-to-text GRU is evaluated across sessions. Native day-specific input layers reproduce low validation phoneme error (9.06% PER), while forcing sessions through non-native input layers increases PER to 22.67–34.43%. Unsupervised moment corrections fail to recover this loss, and small supervised input adapters recover only about 10% of the native-day gap. The strongest practical result is source-layer selection: using a short unlabeled beginning-of-day window to choose a past input layer recovers much of the fixed-source loss, but the immediately previous session is an even stronger simple baseline. A secondary geometry-only analysis of T12 diagnostic blocks shows similar longitudinal structure across spike-power and threshold-crossing features. These results suggest that recency should be the default adaptation prior, while drift geometry is a useful signal for deciding when that default may need to be overridden.


## 1 Introduction

High-performance speech BCIs commonly include session-specific adaptation machinery because intracortical neural features drift across days. This raises a practical question: before retraining a decoder or collecting a long supervised calibration set, how much adaptation is enough?

We examine this question in the official T15 brain-to-text GRU baseline from a rapidly calibrating speech neuroprosthesis study [Card et al., 2024]. The model contains learned day-specific input layers, followed by a GRU and output head. This structure creates a useful stress test: if a target session is decoded through the wrong input layer, performance measures the decoder-facing cost of cross-session input mismatch. We then ask whether simple feature corrections, small learned adapters, or geometry-guided source selection can recover the lost performance.

The contribution is not a new speech decoder. Instead, we provide a decoder-facing adaptation audit:

- Non-native input layers cause a large PER degradation in T15.
- Mean/variance correction is not enough, and small learned adapters give only modest recovery.
- Geometry-guided source selection recovers much of the fixed-source degradation, but previous-session recency is the strongest simple baseline.
- T12 diagnostic blocks show a similar geometry/recency structure in a second dataset, without claiming T12 decoder replication.

We do not claim that neural drift geometry is new. Rather, we show that in a deployed-style brain-to-text decoder with learned day-specific input layers, short-window geometry is actionable for choosing among existing input-layer states and for framing recency as an explicit adaptation prior.

## 2 Related Work

Intracortical communication BCIs have advanced rapidly from handwriting-based brain-to-text decoding [Willett et al., 2021; Fan et al., 2023] to speech neuroprostheses that decode attempted speech into text or spoken output [Willett et al., 2023; Card et al., 2024]. These systems commonly rely on recurrent neural decoders and session-specific adaptation or recalibration to maintain performance as recordings change over time.

Cross-session stability has also been studied through the lens of neural geometry. Manifold-alignment methods can stabilize fixed decoders by aligning low-dimensional neural activity across recording instabilities [Degenhart et al., 2020], and recent dynamics-aware approaches such as NoMAD use temporal structure to perform unsupervised alignment over weeks to months [Karpowicz et al., 2025]. In speech BCI specifically, a recent preprint proposes adversarial session-invariant learning to reduce session-specific information while preserving phoneme decoding structure [Zhang

et al., 2026]. Continual recalibration with pseudo-labels provides a complementary route, using language-model corrections to update communication decoders online [Fan et al., 2023].

Our work is narrower. We do not train a new robust decoder or propose a full recalibration method. Instead, we audit the official T15 decoder to ask which low-cost adaptation rungs are sufficient, when they fail, and how recency and drift geometry can guide source-layer choice.

# 3 Data and Decoder

T15. The main experiments use T15 copy-task neural features from the official brain-to-text dataset [Card et al., 2024]. We evaluate 41 validation sessions with ground-truth phoneme labels. Neural inputs are passed through the official GRU checkpoint and decoded greedily at the phoneme level using CTC collapse.

Official baseline. The decoder contains learned day-specific input layers, a GRU backbone, and an output layer over phoneme classes. In the native-day condition, each session uses its own input layer. In cross-day stress tests, target sessions use a fixed or selected non-native source input layer while the rest of the decoder remains unchanged.

T12 diagnostic blocks. As a secondary check, we analyze diagnostic blocks from the T12 high-performance speech neuroprosthesis dataset [Willett et al., 2023]. This is a geometry-only analysis of spike-power and threshold-crossing feature variants; we do not reproduce the T12 decoder or report T12 PER.

# 4 Methods

## 4.1 Decoder-Facing Metrics

For each T15 validation trial, we compute greedy CTC phoneme predictions and phoneme error rate (PER). We report phoneme-weighted PER unless otherwise stated. Quality diagnostics such as blank rate, confidence, and entropy are saved, but the core results use PER.

## 4.2 Cross-Day Stress Test

We compare native-day decoding against three fixed non-native source layers: early (t15.2023.08.13), middle (t15.2023.11.26), and late (t15.2025.04.13). This tests whether the decoder degradation is tied to one bad source layer or is a general consequence of input-layer mismatch.

## 4.3 Adaptation Ladder

Using the middle source layer, we test no correction, target z-scoring, source moment matching, diagonal affine calibration, and full input-layer calibration. Moment methods use full-session unsupervised feature statistics. Learned adapters use the first $K$ labeled validation trials and are evaluated only on the remaining trials.

Table 1: T15 decoder-facing stress test and adaptation ladder. Fixed-source and moment-correction rows use the middle source t15.2023.11.26. Learned adapter rows use $K = 20$ calibration trials and are evaluated on remaining trials.

| Condition | PER | Recov. | Note |
|---|---|---|---|
| Native-day | 9.06% | 100% | correct layer |
| Fixed early | 28.81% | – | 40/41 harmed |
| Fixed middle | 22.67% | 0% | stress source |
| Fixed late | 34.43% | – | 40/41 harmed |
| Target z-score | 22.75% | -0.6% | fails |
| Moment match | 22.73% | -0.4% | fails |
| Diag. affine, $K = 20$ | 20.21% | 10.1% | adapter |
| Input layer, $K = 20$ | 20.27% | 9.7% | learned layer |

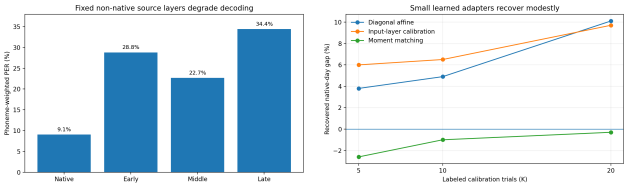


Figure 1: Left: fixed non-native source layers strongly degrade T15 phoneme decoding relative to native-day decoding. Right: small learned calibration recovers only a modest fraction of the fixed-source gap.

## 4.4 Drift Geometry and Source Selection

For each source-target pair, we compute temporal distance, mean and scale shifts, relative covariance shift, CORAL-style distance, and PCA subspace distances. We then select a source input layer by nearest covariance geometry. To approximate deployment, a K-shot variant estimates target geometry from only the first $K \in \{5, 10, 20\}$ unlabeled validation trials and evaluates on the remaining trials.

Because temporal recency is an obvious baseline, we also evaluate the immediately previous input layer. Finally, we test a simple recency-aware override: use the previous input layer by default and switch to the geometry-selected older source only if its covariance distance is less than $\alpha$ times the distance to the previous source.

# 5 Experiments

## 5.1 Input-Layer Mismatch Produces Large Decoder-Facing Degradation

Native-day decoding reaches 9.06% phoneme-weighted PER, confirming that the local evaluation reproduces the expected behavior of the official checkpoint. When all sessions are forced through non-native source input layers, PER rises to 22.67–34.43%, harming 40/41 sessions for each source choice.

## 5.2 Moment Correction Is Not Enough

The natural first hypothesis is that cross-session mismatch is mostly a mean/variance shift. The data do not support this. Target z-scoring and moment matching to the source distribution remain essentially tied with

Table 2: Beginning-of-day source selection on remaining validation trials. Previous-session recency is the strongest simple baseline, while K-shot geometry selection remains close and often selects non-previous sources.

| Method | $K = 5$ | $K = 10$ | $K = 20$ |
|---|---|---|---|
| Native-day | 8.95% | 8.88% | 8.90% |
| Fixed middle | 22.51% | 22.40% | 22.30% |
| Previous source | 13.09% | 12.84% | 12.21% |
| K-shot geometry source | 14.00% | 13.19% | 12.74% |
| Best simple override | 13.09% | 12.90% | 12.18% |

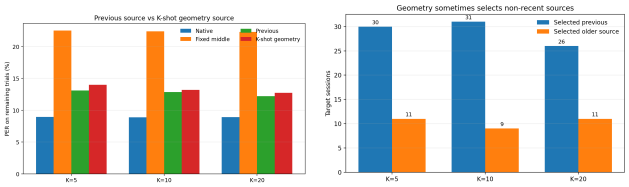


Figure 2: Left: previous-session recency is a strong source-selection baseline and slightly outperforms unconditional K-shot geometry selection. Right: geometry still reveals non-recent source matches, suggesting sessions are not always closest to the immediately previous day.

no correction, and are slightly worse in aggregate. This suggests that the learned day-specific input layers capture transformations richer than channel-wise mean and variance.

## 5.3 Small Learned Adapters Help, But Do Not Solve Drift

Supervised diagonal affine and input-layer calibration improve PER modestly. With $K = 20$, diagonal affine reduces weighted PER from 21.51% to 20.21%, while input-layer calibration gives 20.27%. These correspond to only about 10% of the native-day gap. Thus small learned adapters are a useful rung of the adaptation ladder, but they do not replace the native day-specific layer.

## 5.4 Source Selection Recovers More Than Small Adapters

Geometry-guided source selection gives a much stronger recovery than directly adapting a fixed middle source. In an offline all-candidate setting, choosing the nearest non-native input layer by covariance geometry reaches 11.38% PER. In a stricter past-only setting, it reaches 13.58% PER on 40 sessions, compared with 22.76% for the fixed middle source on the same sessions.

The beginning-of-day version is more realistic. With only the first $K$ unlabeled trials used to estimate target geometry, past-only source selection reaches 14.00%, 13.19%, and 12.74% PER at $K = 5, 10, 20$, recovering 62.8%, 68.1%, and 71.3% of the fixed-source loss. However, the immediately previous input layer is even stronger: 13.09%, 12.84%, and 12.21% PER on the same subsets.

Table 3: Evidence for non-recent source matches. These examples do not prove exact recurrence of neural states, but motivate maintaining a library of past input-layer states rather than only the most recent layer.

| Data | Setting | Evidence | Interpretation |
|---|---|---|---|
| T15 | $K = 5$ | older wins in 3 targets | possible override |
| T15 | $K = 10$ | older wins in 3 targets | consistent across K |
| T15 | $K = 20$ | older wins in 4 targets | older layers useful |
| T12 | diagnostic | 08-13 → 06-21 | 53-day match |
| T12 | diagnostic | 08-25 → 08-13 | older over previous |

## 5.5 Non-Recent Matches Suggest Reusable Input States

The previous-session input layer is the strongest simple baseline, but it is not always the geometrically nearest source. In T15, K-shot geometry selection chooses the previous session in most targets, but selects an older non-previous input layer in 22–30% of cases. Among those non-previous selections, the older source outperforms the previous-session source in 3, 3, and 4 target sessions for $K = 5, 10, 20$, respectively. This suggests that some target sessions may resemble older input-layer states rather than only drifting monotonically away from the previous day.

One possible explanation is that a day-specific input layer may represent a neural recording state rather than a purely chronological day. Changes in recording quality, electrode yield, thresholding, neural strategy, or user state could move a new session closer to an older feature geometry than to the immediately preceding session. In this view, source selection is closer to memory retrieval over past input states than to choosing the most recent date.

We observe the same qualitative pattern in the T12 diagnostic-block geometry check. Using spike-power features, geometry selects an older non-previous session in 7/15 evaluable targets. Some selections jump substantially backward in time, for example 2022-08-13 selecting 2022-06-21, a 53-day lag, and 2022-08-25 selecting 2022-08-13 rather than the immediately preceding session.

The practical implication is that a longitudinal BCI should not keep only the latest input layer. A more robust system could maintain a small library of past input-layer states, use the previous session as the default, and invoke a recency-aware geometry gate only when the beginning-of-day neural geometry strongly suggests that an older state is a better match.

## 5.6 A Bounded Input-State Library Is Usually Enough

We therefore asked how large this past-state library needs to be. For each target session, we restricted K-shot geometry selection to the previous session, the last 3 sessions, the last 5 sessions, or all past sessions. The immediately previous session remains the best aggregate policy, but a small recent library is close to the all-past library. At $K = 20$, last-5 and all-past selection both reach 12.74% PER, while last-3 gives 12.76%. Expand-

Table 4: K-shot input-state library-size ablation. PER is measured on remaining validation trials after using the first $K$ trials to estimate target geometry. A bounded recent library nearly matches all-past selection, while previous-only recency remains the strongest aggregate baseline.

| $K$ | Previous | Last 3 | Last 5 | All past |
|---|---|---|---|---|
| 5 | 13.09% | 13.47% | 13.55% | 14.00% |
| 10 | 12.84% | 13.19% | 13.19% | 13.19% |
| 20 | 12.21% | 12.76% | 12.74% | 12.74% |

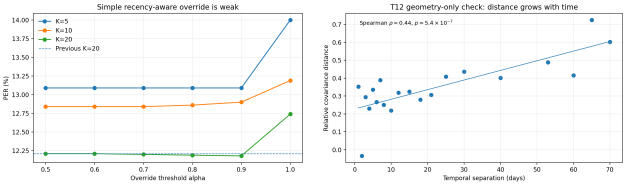


Figure 3: Left: a naive recency-aware geometry override does not substantially beat the previous-session baseline. Right: T12 diagnostic-block covariance distance increases with temporal separation, supporting the broader geometry framing outside T15.

ing the nearest-source search to all past layers does not improve aggregate PER over recency under this simple distance rule; rather, it increases the need for a confidence gate before using older states.

### 5.7 A Single Distance-Ratio Override Is Too Weak

Since geometry sometimes chooses older sources, we tested whether a simple distance-ratio gate can improve over recency: keep the previous source unless the geometry-selected source is at least an $\alpha$ fraction closer in covariance distance. This rule does not yet produce a meaningful improvement. At $\alpha = 0.9$, it ties previous at $K = 5$, is slightly worse at $K = 10$ (12.90% vs 12.84%), and improves only marginally at $K = 20$ (12.18% vs 12.21%). At $\alpha = 1.0$, it approaches unconditional geometry selection and becomes worse. Geometry is therefore a useful candidate signal, but a deployable gate needs richer confidence features.

### 5.8 T12 Diagnostic Blocks Support the Geometry/Recency Framing

T12 diagnostic blocks provide a secondary geometry-only check. Using spike-power features, covariance distance increases with temporal separation across 120 past-source pairs (Spearman $\rho = 0.44$, $p = 5.4 \times 10^{-7}$). Geometry selects the immediately previous session in 8/15 target sessions and an older source in 7/15. Some selections jump far back in time, such as a 53-day lag. This pattern is not limited to spike power: threshold-crossing variants tx1–tx4 also show positive days-distance correlations ($\rho = 0.34$–$0.47$, all $p < 1.3 \times 10^{-4}$) and older-source selections in 6–9 of 15 targets. This supports the idea that recency is important but not sufficient to describe longitudinal neural geometry. Because we do not run the T12 decoder, this result is only supporting evidence, not decoder replication.

## 6 Discussion

The main lesson is that cross-session adaptation should not be treated as a single fixed correction. The native T15 input layers are highly effective, and using the wrong input layer causes large decoder-facing degradation. But simple moment alignment does not emulate those layers, and small learned adapters recover only a small fraction of the lost performance.

The most useful low-cost intervention is not a new adapter, but a better choice of existing input layer. A short unlabeled beginning-of-day window is enough to choose a past layer that recovers much of the fixed-source loss. Yet the strongest simple policy is often even simpler: use the previous session. This changes the adaptation ladder. Recency should be the default prior; geometry should be used as an additional signal for detecting when the previous session is not the right source. This also suggests a practical memory mechanism: instead of overwriting old session adapters, the system should retain a library of past input-layer states and learn when the current session has returned to one of them.

The failed distance-ratio override is also informative. It shows that geometry is not automatically a better policy than recency. A future gate should likely use multiple confidence features: distance margin, previous-source rank, temporal gap, session quality, decoder confidence, and perhaps hidden-state or logit-space diagnostics.

A natural extension is to test whether the input-state library view applies not only across days, but also across speech behaviors. Recent inner-speech recordings from T12, T15, T16, and T17 include attempted, imagined or inner, listening, and silent-reading tasks, and show that inner speech is represented in motor cortex with a motor-intent dimension separating attempted and inner speech [Kunz et al., 2025]. These data could test whether a new neural geometry reflects session drift, speech-mode shift, or both. More broadly, future input-state libraries may need to index both recording state and intended speech mode.

## 7 Limitations

This study is a decoder-facing audit, not a new decoding architecture. Greedy PER is lighter than full WER and avoids the language-model/Redis evaluation stack. T12 is geometry-only and should not be interpreted as T12 PER replication. The learned adapter and K-shot source-selection experiments simulate a beginning-of-day calibration window within the validation split and evaluate only on the remaining trials. They are intended to map adaptation rungs, not to optimize a final deployment benchmark.

## 8 Conclusion

For T15 brain-to-text decoding, fixed non-native input layers break the decoder, moment correction is too weak, and small learned adapters recover only modestly. Most of the practical recovery comes from selecting a compatible past input layer. The immediately previous session is a strong default, while drift geometry is a measurable signal for deciding when that default may need to be overridden. This supports a recency-aware, geometry-guided adaptation ladder: start with the previous input state, measure drift geometry from a short unlabeled window, and move to stronger calibration only when the geometry indicates that simple source selection is not enough.

## Ethical Statement

This work uses publicly released intracortical BCI datasets and reports only aggregate analyses. It does not attempt to infer private inner speech or deploy a decoder. Future extensions to inner-speech or cross-behavior state selection should include explicit intent-gating and privacy safeguards.

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